方法：菲涅尔衍射积分．考纳螺线．

周期性物体的菲涅尔衍射

1. 一维余弦光栅.

$$t(x_0) = \frac{1}{2} + \frac{\beta}{2}\cos(2\pi\xi_0 x_0).$$

用单位单色平面波垂直照射.

$$U_0'(x_0) = t(x_0).$$

代入菲涅尔衍射公式(4.4.4). 距光栅 d 处平面的衍射光场:

$$U_i(x_i) = \frac{e^{ikd}}{i\lambda d}\iint_{-\infty}^{\infty} t(x_0)\, e^{\frac{ik}{2d}(x_i - x_0)^2}\, dx_0$$

$$= \frac{e^{ikd}}{i\lambda d}\, t(x_0) * e^{\frac{ik}{2d}x_i^2}$$

两边同时FT.

$$F\{U_i(x_i)\} = \frac{e^{ikd}}{i\lambda d}\, F\{t(x_0)\} \cdot F\{e^{\frac{ik}{2d}x_i^2}\}$$

又由第2章:

$$\begin{cases} F\{t(x_0)\} = \frac{1}{2}\delta(\xi) + \frac{\beta}{4}\delta(\xi - \xi_0) + \frac{\beta}{4}\delta(\xi + \xi_0) \\ F\{e^{\frac{ik}{2d}x_i^2}\} = i\lambda d\, e^{-i\pi\lambda d\xi^2} \end{cases}$$

$$\therefore \quad F\{U_i(x_i)\} = \frac{e^{ikd}}{i\lambda d}\left[\frac{1}{2}\delta(\xi) + \frac{\beta}{4}\delta(\xi - \xi_0) + \frac{\beta}{4}\delta(\xi + \xi_0)\right] i\lambda d\, e^{-i\pi\lambda d\xi^2}$$

$$= e^{ikd}\left[\frac{1}{2}\delta(\xi) + \frac{\beta}{4}e^{-i\pi\lambda d\xi_0^2}\delta(\xi - \xi_0) + \frac{\beta}{4}e^{-i\pi\lambda d\xi_0^2}\delta(\xi + \xi_0)\right]$$

作 IFT:

$$U_i(x_i) = \underbrace{e^{ikd}}_{\text{常相位因子}}\left[\frac{1}{2} + \frac{\beta}{2}\underbrace{e^{-i\pi\lambda d\xi_0^2}}_{\text{调制相位因子}}\cos(2\pi\xi_0 x_i)\right].$$

除了常相位因子与调制相位因子与余弦光栅 $t(x_0) = \frac{1}{2} + \frac{\beta}{2}\cos(2\pi\xi_0 x_0)$ 一致.

衍射光强：

$$I_i = |U_i(x_i)|^2 = \left|\frac{1}{2} + \frac{\beta}{2} e^{-i\pi\lambda d\xi_0^2} \cos(2\pi\xi_0 x_i)\right|^2$$

讨论调制相位因子：

① $e^{i\pi\lambda d\xi_0^2} = e^{-i\pi 2m} = 1$ ， $m = \pm 1, \pm 2, \cdots$

即

$$d = \frac{2mL_0^2}{\lambda} = m d_T \quad \left(L_0 = \frac{1}{\xi_0}\right)$$

L_0 → 光栅的周期.

d_T ← 塔尔博特距离

衍射光强

$$I_i = \left|\frac{1}{2} + \frac{\beta}{2} \cos(2\pi\xi_0 x_i)\right|^2$$

观测的像为余弦光栅的再现. —— 塔尔博特像.

② $e^{-i\pi\lambda d\xi_0^2} = e^{-i\pi(2m+1)} = -1$， $m = \pm 1, \pm 2, \cdots$.

即：

$$d = \frac{(2m+1)}{\lambda} L_0^2 = \left(m + \frac{1}{2}\right) d_T.$$

衍射光强：

$$I_i = \left|\frac{1}{2} - \frac{\beta}{2} \cos(2\pi\xi_0 x_i)\right|^2$$

可见度反转， 负塔尔博特像.

③ $e^{-i\pi\lambda d\xi_0^2} = e^{-i\pi\frac{2m-1}{2}} = -i \quad m=\pm1, \pm2, \cdots$.

$$d = (m-\tfrac{1}{2})\frac{L^2}{\lambda} = (\tfrac{m}{2}-\tfrac{1}{4})d_T.$$

衍射光强：

$$I_i = \left|\tfrac{1}{2} + i\tfrac{g}{2}\cos(2\pi\xi_0 x_i)\right|^2 = 1 + \frac{g^2}{4}\cos^2(2\pi\xi_0 x_i)$$

$$= \tfrac{1}{4} + \frac{g^2}{8} + \frac{g^2}{8}\cos[2\pi(2\xi_0)x_i]$$

相比塔尔博特像频率翻倍. —— 负塔尔博特子像.

2. 矩形光栅

周期为 L_0，振幅透射率：

$$t(x_0) = \begin{cases} 1 & |x_0 - 2nL_0| \le \frac{L_0}{2} \\ 0 & |x_0 - 2nL_0| > \frac{L_0}{2} \end{cases}$$

Fourier series:

$$t(x_0) = \sum_{n=0}^{\infty} A_n \cos(2\pi n x_0 / L_0).$$

若只取前 2 项，$A_0 = \frac{1}{2}$，$A_1 = \frac{1}{\pi}$ $\longrightarrow$ 余弱光栅.

若用单位平面波垂直入射.

$$U_0'(x_0, z=0) = \sum_{n=0}^{\infty} \frac{A_n}{2} \cos(2\pi n x_0 / L_0)$$

FT:

$$A_0(\xi, z=0) = \sum_{n=0}^{\infty} \frac{A_n}{2} [\delta(\xi + \frac{n}{L_0}) + \delta(\xi - \frac{n}{L_0})].$$

经距离 d 传播，由 (4.3.31) 在菲涅尔近似下传递函数，可得其频谱.

作 IFT. 有:

$$U_i(x_i, z=d) = \sum_{n=0}^{\infty} A_n \cos(\frac{2\pi n x_i}{d}) e^{-i\pi\lambda d (\frac{n}{L_0})^2}$$

若取

$$\pi\lambda d (\frac{n}{L_0})^2 = 2m\pi, \quad m = 1, 2, \cdots.$$

当距离 d 满足:

$$d = \frac{2mL_0^2}{n^2\lambda} = \frac{m}{n} d_T$$

时，有: $U_i(x_i) = t(x_0)$. —— 自成像.

3. 任意形状周期函数的菲涅尔衍射

一般周期函数可表为单个非周期函数与梳状函数卷积.

$$t(x_0) = t_0(x_0) * \sum_{n=-\infty}^{\infty} \delta(x - nL_0) \quad (n=0, \pm1, \pm2, \cdots)$$

用波长 λ 单位振幅单色平面波垂直照射. $U_0(x_0)=1$.

$$U_0'(x_0) = t(x_0)\,U_0(x_0) = t(x_0) = \sim.$$

FT:

$$A_0'(\xi) = T_0(\xi) \sum_{n=-\infty}^{\infty} \delta(\xi - n/L_0)$$

由 (4.3.32).

$$A_i'(\xi) = H(\xi)\,A_0'(\xi).$$

由 (4.4.11). 菲涅尔近似下:

$$H(\xi) = e^{ikd}\, e^{-i\pi\lambda d\xi^2}$$

代入:

$$A_i(\xi) = e^{ikd} e^{-i\pi\lambda d\xi^2} T_0(\xi) \sum_{n=-\infty}^{\infty} \delta(\xi - \frac{n}{L_0}) = T_0(\xi)\, e^{ikd} \sum_{n=-\infty}^{\infty} \delta(\xi_0 - \frac{n}{L_0})\, e^{-i\pi\lambda d\xi^2}$$

IFT:

$$U_i(x_i) = t_0(x_i)\, e^{ikd} * F\left\{ \sum_{n=-\infty}^{\infty} \delta(\xi - \frac{n}{L_0})\, e^{-i\pi\lambda d\xi^2} \right\}$$

$$= t_0(x_i)\, e^{ikd} * \left[\sum_{n=-\infty}^{\infty} e^{i2\pi x_i \frac{n}{L_0}}\, e^{-i\pi\lambda d(\frac{n}{L_0})^2} \right].$$

对频率为 $\frac{n}{L_0}$ 分量平面波，在观察平面仅引起了 $e^{ikd}\, e^{-i\pi\lambda d(\frac{n}{L_0})^2}$ 的相移.

若有: $\pi\lambda d(\frac{n}{L_0})^2 = 2m\pi \iff d = \frac{2mL_0^2}{n^2\lambda} = \frac{m}{n^2} d_T$. $m=1, 2, \cdots$.

塔尔博特距离.

则: $e^{i\sim} = 1$.

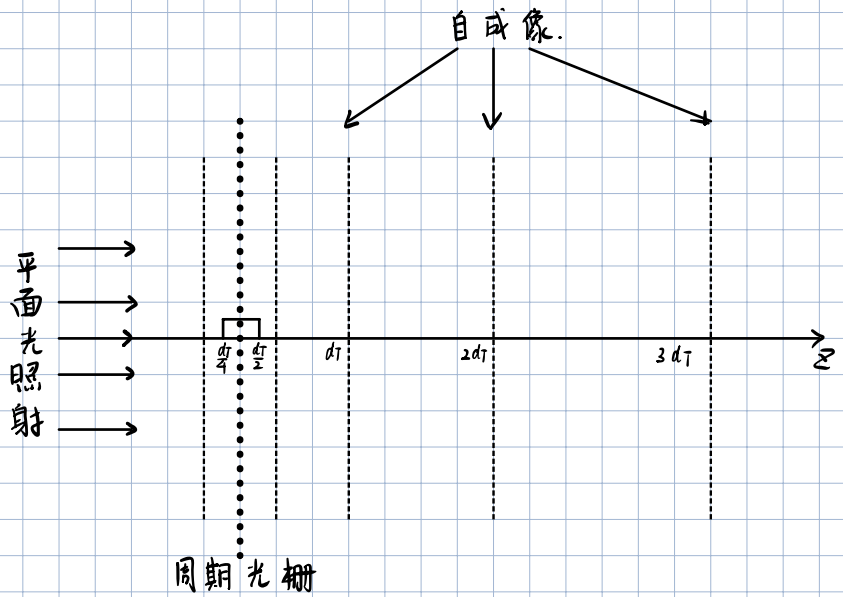
即: 不同频率 $\frac{n}{L_0}$ 成份在观察平面上引起相移除了一常数因子外，都是 2π 整数倍.

此时： $\sum_{n=-\infty}^{\infty} e^{i2\pi x_i \frac{n}{L_0}} = \sum_{n=-\infty}^{\infty} \delta(x_i - nL_0)$.

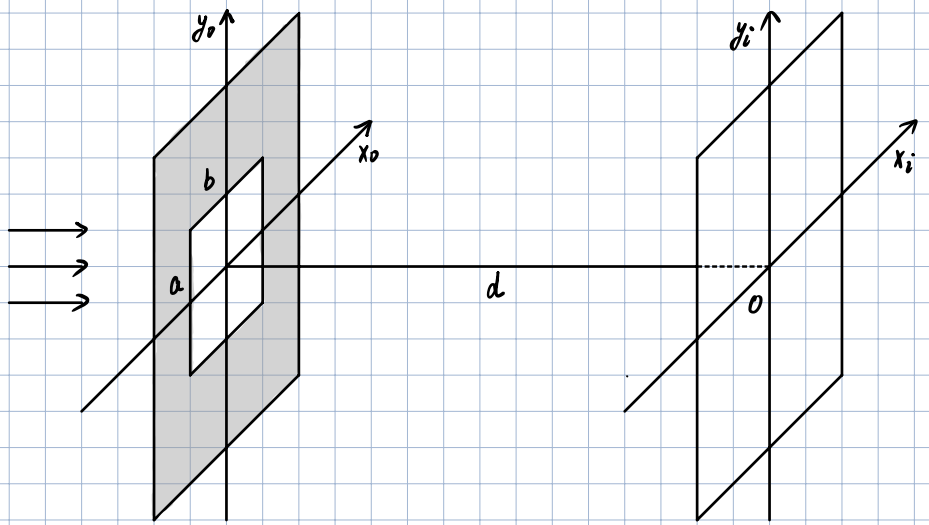
$$U_i(x_i) = t_0(x_i) e^{ikd} * \sum_{n=-\infty}^{\infty} \delta(x_i - nL_0) = t(x_i) e^{ikd}.$$

$$I(x_i) = |U_i(x_i)|^2 = |t(x_i)|^2$$ 物体像.

conclussion： $\forall$ 形状周期性物体都有塔尔博特现象.



矩形孔菲涅尔衍射计算



$$t_0(x_0, y_0) = rect\left(\frac{x_0}{a}, \frac{y_0}{b}\right)$$

$$U_0'(x_0, y_0) = t(x_0, y_0)\, U_0(x_0, y_0) = rect\left(\frac{x_0}{a}, \frac{y_0}{b}\right)$$

$$U_i'(x_i, y_i) = \frac{e^{ikd}}{i\lambda d} \iint_{-\infty}^{\infty} t(x_0, y_0)\, e^{\frac{ik}{2d}\left[(x_i - x_0)^2 + (y_i - y_0)^2\right]} dx_0\, dy_0$$

$$= \frac{e^{ikd}}{i\lambda d} \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{\frac{ik}{2d}(x_i - x_0)^2} dx_0 \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{\frac{ik}{2d}(y_i - y_0)^2} dy_0$$

$$= \frac{e^{ikd}}{i} U_{ix}(x_i)\, U_{iy}(y_i)$$

其中，

$$U_{ix}(x_i) = \frac{1}{\sqrt{\lambda d}} \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{\frac{ik}{2d}(x_i - x_0)^2} dx_0$$

$$U_{iy}(y_i) = \frac{1}{\sqrt{\lambda d}} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{\frac{ik}{2d}(y_i - y_0)} dy_0$$

上面积分是复杂的，要应用考纽螺线图解法.

作变量替换：

$$x'=\sqrt{\frac{k}{\pi d}}(x_i-x_0),\quad x_1'=\sqrt{\frac{k}{\pi d}}(\frac{a}{2}-x_0),\quad x_2'=-\sqrt{\frac{k}{\pi d}}(\frac{a}{2}+x_0).$$

$$y'=\sqrt{\frac{k}{\pi d}}(y_i-y_0),\quad y_1'=\sqrt{\frac{k}{\pi d}}(\frac{b}{2}-y_0).\quad y_2'=-\sqrt{\frac{k}{\pi d}}(\frac{b}{2}+y_0).$$

$$U_i(x_i,y_i)=\frac{e^{ikz}}{i\lambda z}\int_{x_1'}^{x_2'}e^{\frac{i\pi}{2}x'^2}dx'\int_{y_1'}^{y_2'}e^{\frac{i\pi}{2}y'^2}dy'$$

$$=\frac{e^{ikz}}{i}U_{ix}(x_i)U_{iy}(y_i)$$

$$U_{ix}(x_i)=\frac{1}{\sqrt{2}}\int_{x_1'}^{x_2'}e^{\frac{i\pi}{2}x'^2}dx',\quad U_{iy}(y_i)=\frac{1}{\sqrt{2}}\int_{y_1'}^{y_2'}e^{\frac{i\pi}{2}y'^2}dy'$$

由欧拉公式：

$$e^{\frac{i\pi}{2}x'^2}=\cos(\frac{\pi}{2}x'^2)+i\sin(\frac{\pi}{2}x'^2),\quad e^{\frac{i\pi}{2}y'^2}=\cos(\frac{\pi}{2}y'^2)+i\sin(\frac{\pi}{2}y'^2)$$

$$U_{ix}(x_i)=\frac{1}{\sqrt{2}}\int_{x_1'}^{x_2'}e^{\frac{i\pi}{2}x'^2}dx'=\frac{1}{\sqrt{2}}\left(\int_0^{x_2'}e^{\frac{i\pi}{2}x'^2}dx'-\int_{x_1'}^{0}e^{\frac{i\pi}{2}x'^2}dx'\right).$$

$$=\frac{1}{\sqrt{2}}[C(x_2')-C(x_1')]+\frac{i}{\sqrt{2}}[S(x_2')-S(x_1')].$$

$$U_{iy}(y_i)=\frac{1}{\sqrt{2}}[C(y_2')-C(y_1')]+\frac{i}{\sqrt{2}}[S(y_2')-S(y_1')].$$

其中，$C(\alpha)$，$S(\alpha)$ 为菲涅尔函数.

$$\begin{cases}C(\alpha)=\int_0^{-\alpha}\cos(\frac{\pi x^2}{2})dx\\ S(\alpha)=\int_0^{-\alpha}\sin(\frac{\pi x^2}{2})dx\end{cases}.$$

$$I_i=|U_i|^2.$$

特殊矩形孔菲涅尔衍射

1. 自由空间.

$$a = b = \infty.$$

$$x_1' = y_1' = -\infty. \quad x_2' = y_1' = -\infty.$$

$$U_i(x_i, y_i) = \frac{1}{2i} e^{ikd}(1+i)(1+i) = e^{ikd}$$

2. 直边衍射.

衍射屏为半无穷大平面.

$$x_1' = -\infty \quad x_2' = \infty. \quad y_1' = -\sqrt{\frac{2}{\lambda d}} y_i \quad y_i' = \infty$$

$$U_i(x_i, y_i) = \frac{e^{ikd}}{2i}(1+i)\left\{\left[\frac{1}{2} - C(y_i')\right] + i\left[\frac{1}{2} - S(y_i')\right]\right\}$$

衍射光场仅与观察点坐标 y_i 有关.

$$I_i(x_i, y_i) = |U_i(x_i, y_i)|^2 = \frac{1}{2}\left\{\left[\frac{1}{2} - C(y_i')\right]^2 + \left[\frac{1}{2} - S(y_i')\right]^2\right\}.$$

对观察点进行讨论.

(1) 观察点P位于半无穷大平面几何阴影边缘.

$$y_i' = 0.$$

$$U_i(x_i, 0) = \frac{e^{ikd}}{2} \frac{1}{2}(1+i)^2 = \frac{1}{2} e^{ikd}$$

$$I_i(x_i, 0) = \frac{1}{4}$$

(2) 观察点M在阴影区，且离直边L不远.

$$y_i<0,\quad y_i'=-\sqrt{\frac{2}{\lambda d}}$$

考纽线上与M点相应于 $x'=y_i'$，C点相应于 $x'\to\infty$. 此时有：

$$\left[\frac{1}{2}-C(y_i')\right]+i\left[\frac{1}{2}-S(y_i')\right]=\overline{MC}\,e^{i\theta}$$

其中，θ 为 $\overline{MC}$ 与横轴夹角. 代入：

$$I_i(M)=\frac{1}{2}\overline{MC}$$

(3) 观察点由M移至阴影区深处D，y_i ↗. M点沿考纽螺线向C点移动，$\overline{MC}$ ↓. 由上式知 I ↓.

考虑由D继续向 y_i 负方向移动. $y_i\to-\infty$，$y_i'\to\infty$. M、C重合.

$$U_i(x_i,\infty)=0,\quad I_i(x_i,\infty)=0.$$

(4) 观察点由几何阴影区边缘移至照明区L点，$y_i>0$，$y_i'=-\sqrt{\frac{2}{\lambda d}}$

在考纽螺线上与L对应点位于第三象限.

$$I_i(L)=\frac{1}{2}\overline{LC}^2.$$

当 y_i 由极小值逐渐加大，y_i' 的负值绝对值也↗. $\overline{LC}$ 变长到L点达到Q点，Q点处 $y_i'\approx1.22$. I max. $I_i\approx1.37$.

当 y_i' 由 -1.22 继续减小至R点，R处 $y_i'\approx-1.87$.

∴ 若在照明区域观察点由R点继续沿 y_i 正向移动，观测点相对强度 I_i/I_{io} 在1附近振荡，且振幅↓.

当 $y' \to \infty$. 相当于位于照明区且远离阴影边缘 B. 光强

$$I_i(CB) = \frac{1}{2}\overline{BC}^2 = I_{i0}$$

3. 长狭缝菲涅尔衍射.

矩形孔某边 $a\to\infty$ 或 $b\to\infty$，另一边为有限值.

$$x_1'=-\infty.\quad x_2'=\infty.\quad y_1'=-\sqrt{\frac{2}{\lambda d}}(b+y_i)\quad y_2'=-\sqrt{\frac{2}{\lambda d}}(b-y_i)$$

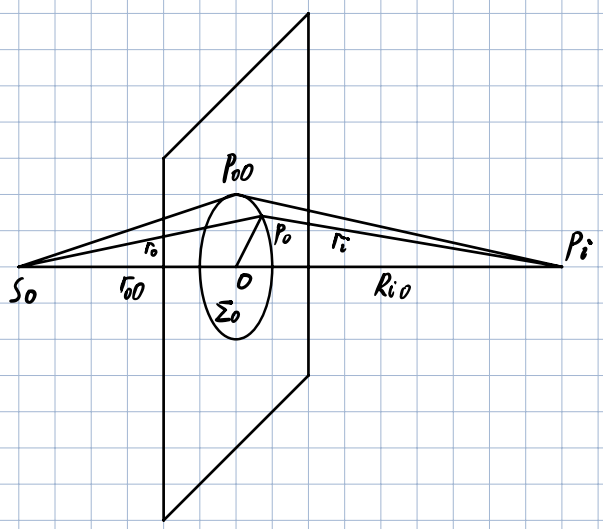
$$U_i(x_i,y_i)=\frac{e^{ikzd}}{2i}(1+i)\left\{[C(y_2')-C(y_1')]+i[S(y_2')-S(y_1')]\right\}.$$

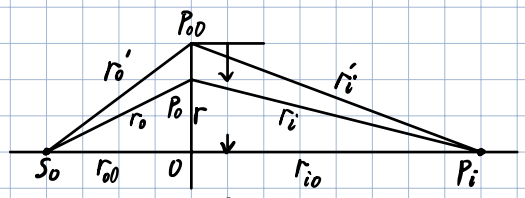
令 $y_0'=y_2'-y_1'=b\sqrt{\frac{2}{\lambda d}}$. 对给定观测点，y_0' 为一无量纲常数，反映了孔径尺寸对孔径间距离的相对大小. 如此，光强：

$$I_i(x_i,y_i)=\frac{1}{2}(1+i)[(C(y_1'+y_0')-C(y_1'))]^2+[S(y_1'+y_0')-S(y_1')]^2$$

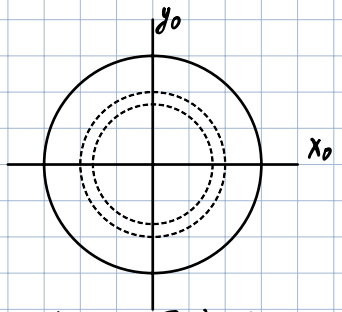
- 对较大 y_0'，相当于2个直边衍射组合.
- 对较小 y_0'，趋于夫琅和费衍射.

圆孔菲涅尔衍射





子午面



菲涅尔半周期带.

若 P_0 为孔径 Σ_0 内任一点，由基尔霍夫衍射公式，P_i 点光场复振幅：

$$U_i(P_i) = \frac{1}{i\lambda}\iint_{\Sigma_0} U_0(P_0)\frac{e^{ikr}}{r}\cos\theta\, ds_0$$

其中，

$$U_0(P_0) = A_0\frac{e^{ikr_0}}{r_0}$$

假定 $a \ll r_0$，$\Rightarrow \cos\theta_0 \approx 1$.

$$U_i(P_i) = \frac{1}{i\lambda}\iint_{\Sigma_0} A_0\frac{e^{ikr_0}}{r_0}\frac{e^{ikr_i}}{r_i}\,ds = \frac{A_0}{i\lambda}\iint_{\Sigma_0}\frac{e^{ik(r_0+r_i)}}{r_0 r_i}\,ds$$

极坐标下，

$$U_i(P_i) = \frac{A_0}{i\lambda}\int_0^{2\pi} d\varphi\int_0^a r\frac{e^{ik(r_0+r_i)}}{r_0 r_i}\,dr_0$$

由 $r^2 = r_0^2 - r_{00}^2 = r_i^2 - r_{i0}^2 \Rightarrow r\,dr = r_0\,dr_0 = r_i\,dr_i$.

$\therefore$

$$d(r_i + r_0) = \left(\frac{1}{r_i} + \frac{1}{r_0}\right) r\,dr$$

令 $\alpha = r_i + r_0$，$\alpha_0 = r_{i0} + r_{00}$，$\alpha_1 = \sqrt{r_{i0}^2 + a^2} + \sqrt{r_{00}^2 + a^2}$，其中，$\alpha_0$，$\alpha_1$ 分别为光由 S_0 经孔径 Σ_0 达 P_i 间最大与最小光程.

$$U_i(P_i) = \frac{2\pi A_0}{i\lambda}\int_{\alpha_0}^{\alpha_1}\frac{e^{ik\alpha}}{\alpha}\,d\alpha.$$

在 $r_i + r_0 \ll a$ 条件下，上式积分分母 α 可由 r_{00} 代替.

$$U_i(P_i) \approx \frac{2\pi A_0}{i\lambda r_{00}}\int_{\alpha_0}^{\alpha_1} e^{ik\alpha}\,d\alpha = U_{i0}(P_i)(1-e^{i\pi m}).$$

其中，

$$U_{i0}(P_i) = A_0\frac{e^{ik(r_{00}+r_{i0})}}{r_{00}+r_{i0}}$$

为不存在屏时 P_i 点光场，

$$\frac{m\lambda}{2} = \alpha_1 - \alpha_0$$

表示由 S_0 到 P_i 点透过孔径 Σ_0 传播的最大光程与最小光程间差值.

光强，

$$I_i(P_i) = 4I_{i0}(P_i)\sin^2\frac{m\pi}{2}$$

$$I_{i0}(P_i) = |U_{i0}(P_i)|^2$$

由上式，

m为偶，$I_i = 0$；

m为奇，I_i 极大 $4I_{i0}$ 屏后面的轴上出现亮斑.

考虑 S_0 与 P_i 位置将圆孔分割为若干同心圆环，每圆环构成一个带，第 n 个圆半径 r_n 有：

$$(S_0P_{0n} + P_{0n}P_i) - (S_0O + OP_i) = \frac{n\lambda}{2}.$$

由几何关系：

$$\sqrt{r_{i0}^2 + a^2} + \sqrt{r_{00}^2 + a^2} - (r_{i0} + r_{00}) = \frac{n\lambda}{2}$$